

```

(kali㉿kali)-[~]
$ mkdir -p ~/CIP-B103-Lab3/{evidence,working,exported,reports,screenshots,scripts}
cd ~/CIP-B103-Lab3

(kali㉿kali)-[~/CIP-B103-Lab3]
$ pwd
/home/kali/CIP-B103-Lab3

(kali㉿kali)-[~/CIP-B103-Lab3]
$ find . -maxdepth 1 -type d -print
.
./working
./screenshots
./evidence
./scripts
./reports
./exported

(kali㉿kali)-[~/CIP-B103-Lab3]
$ tree ~/CIP-B103-Lab3
/home/kali/CIP-B103-Lab3
├── evidence
├── exported
├── reports
├── screenshots
├── scripts
├── working
└──

7 directories, 0 files

(kali㉿kali)-[~/CIP-B103-Lab3]
$

```

```

(kali㉿kali)-[~/CIP-B103-Lab3]
$ sudo apt update
[sudo] password for kali:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1734 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali㉿kali)-[~/CIP-B103-Lab3]
$ sudo apt install -y apache2 tshark wireshark python3-scapy
apache2 is already the newest version (2.4.68-1).
apache2 set to manually installed.
tshark is already the newest version (4.6.6-1).
tshark set to manually installed.
wireshark is already the newest version (4.6.6-1).
wireshark set to manually installed.
python3-scapy is already the newest version (2.7.0+dfsg1-1).
python3-scapy set to manually installed.
The following packages were automatically installed and are no longer required:
 libraw23t64 libvidstab1.1 openjdk-21-jre openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1734

(kali㉿kali)-[~/CIP-B103-Lab3]
$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2

(kali㉿kali)-[~/CIP-B103-Lab3]
$ systemctl status apache2
● apache2.service - The Apache HTTP Server

```

```
(kali㉿kali)-[~/CIP-B103-Lab3]
└─$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-07-26 09:00:32 EDT; 1h 53min ago
     Invocation: 2fcc79af483740eda369fc8dc55f0709
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 949 (apache2)
      Status: "Total requests: 10; Idle/Busy workers 100/0;Requests/sec: 0.00147; Bytes served/sec: 19 B/sec"
        Tasks: 7 (limit: 2108)
       Memory: 20.8M (peak: 37.2M)
          CPU: 1.078s
    CGroup: /system.slice/apache2.service
            └─949 /usr/sbin/apache2 -k start -DFOREGROUND
              └─996 /usr/sbin/apache2 -k start -DFOREGROUND
                └─997 /usr/sbin/apache2 -k start -DFOREGROUND
                  └─998 /usr/sbin/apache2 -k start -DFOREGROUND
                    └─999 /usr/sbin/apache2 -k start -DFOREGROUND
                      └─1000 /usr/sbin/apache2 -k start -DFOREGROUND
                        └─8928 /usr/sbin/apache2 -k start -DFOREGROUND

Jul 26 09:00:32 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jul 26 09:00:32 kali apachectl[949]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.>
Jul 26 09:00:32 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
```

```
(kali㉿kali)-[~/CIP-B103-Lab3]
└─$ wget -O evidence/mySYNFloodCapture.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap'
--2026-07-26 10:54:48-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood
/mySYNFloodCapture.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 260716 (255K) [application/octet-stream]
Saving to: 'evidence/mySYNFloodCapture.pcap'

evidence/mySYNFloodCapture.pcap 100%[=====>] 254.61K 1.18MB/s in 0.2s

2026-07-26 10:54:49 (1.18 MB/s) - 'evidence/mySYNFloodCapture.pcap' saved [260716/260716]

(kali㉿kali)-[~/CIP-B103-Lab3]
└─$ ls -lh evidence/
total 256K
-rw-rw-r-- 1 kali kali 255K Jul 26 10:54 mySYNFloodCapture.pcap

(kali㉿kali)-[~/CIP-B103-Lab3]
└─$ sha256sum evidence/mySYNFloodCapture.pcap | tee reports/syn_capture_sha256.txt
14765b029a72e9c41dd8b4d32f5b1d2c7d9efee0f084949151f183a39baa55f5 evidence/mySYNFloodCapture.pcap

(kali㉿kali)-[~/CIP-B103-Lab3]
└─$
```

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ sudo tshark -i lo -f 'tcp port 80' -a duration:20 -w /tmp/normal_http.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'Loopback: lo'
```

File System

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ sudo tshark -i lo -f 'tcp port 80' -a duration:20 -w /tmp/normal_
http.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'Loopback: lo'
10
```

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ sudo cp /tmp/normal_http.pcapng evidence/
sudo chown kali:kali evidence/normal_http.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ ls -lh evidence/normal_http.pcapng
-rw----- 1 kali kali 13K Jul 26 11:26 evidence/normal_http.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ tshark -r evidence/normal_http.pcapng -Y 'tcp.flags.syn==1 || tcp.flags.fin==1'
-T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e
tcp.flags \
| tee reports/normal_handshake_flags.tsv
1      2026-07-26T11:25:24.030379939-0400      127.0.0.1      40198      127.0.0.1
      80      0x0002
2      2026-07-26T11:25:24.030391815-0400      127.0.0.1      80      127.0.0.1
      40198      0x0012
8      2026-07-26T11:25:24.031857171-0400      127.0.0.1      40198      127.0.0.1
      80      0x0011
9      2026-07-26T11:25:24.031970109-0400      127.0.0.1      80      127.0.0.1
      40198      0x0011
```

zsh: corrupt history file /home/kali/.zsh_history

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ curl --no-keepalive http://127.0.0.1/ >/dev/null
% Total    % Received % Xferd  Average Speed   Time    Time     Time    Current
           %             %             Dload  Upload  Total   Spent    Left     Speed
100  10703  100  10703    0     0    6.25M    0      0      0      0      0      0      0
(kali㉿kali)-[~/CIP-B103-Lab3]
$
```

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ cd ~/CIP-B103-Lab3

(kali㉿kali)-[~/CIP-B103-Lab3]
$ cat > scripts/syn_probe_lab.py <<'PYLAB'
from scapy.all import IP, TCP, RandShort, send

TARGET = '127.0.0.1'
PORT = 80
COUNT = 4

packets = [IP(dst=TARGET)/TCP(sport=RandShort(), dport=PORT, flags='S') for _ in ra
send(packets, verbose=False)
print(f'Sent {COUNT} authorized training SYN packets to {TARGET}:{PORT}')
PYLAB

(kali㉿kali)-[~/CIP-B103-Lab3]
$ ls -l scripts/
total 4
-rw-rw-r-- 1 kali kali 288 Jul 26 11:30 syn_probe_lab.py
```

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ sudo tshark -i lo -f 'tcp port 80' -c 20 -w /tmp/bounded_syn_activity.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'Loopback: lo'
12 ^C

(kali㉿kali)-[~/CIP-B103-Lab3]
$ sudo cp /tmp/bounded_syn_activity.pcapng evidence/

(kali㉿kali)-[~/CIP-B103-Lab3]
$ sudo chown kali:kali evidence/bounded_syn_activity.pcapng

(kali㉿kali)-[~/CIP-B103-Lab3]
$ ls -lh evidence/
total 276K
-rw----- 1 kali kali 1.4K Jul 26 11:34 bounded_syn_activity.pcapng
-rw-rw-r-- 1 kali kali 255K Jul 26 10:54 mySYNFloodCapture.pcap
-rw----- 1 kali kali 13K Jul 26 11:26 normal_http.pcapng

(kali㉿kali)-[~/CIP-B103-Lab3]
$ tshark -r evidence/bounded_syn_activity.pcapng -Y 'tcp.flags.syn==1' -T fields
-e frame.number -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags \
| tee reports/bounded_syn_packets.tsv
1      127.0.0.1      51631  127.0.0.1      80      0x0002
2      127.0.0.1      80     127.0.0.1      51631   0x0012
4      127.0.0.1      45323  127.0.0.1      80      0x0002
5      127.0.0.1      80     127.0.0.1      45323   0x0012
7      127.0.0.1      9145   127.0.0.1      80      0x0002
8      127.0.0.1      80     127.0.0.1      9145    0x0012
10     127.0.0.1      33581  127.0.0.1      80      0x0002
11     127.0.0.1      80     127.0.0.1      33581   0x0012

(kali㉿kali)-[~/CIP-B103-Lab3]
$
```



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~/CIP-B103-Lab3]
$ sudo python3 scripts/syn_probe_lab.py
[sudo] password for kali:
Sent 4 authorized training SYN packets to 127.0.0.1:80
```

```
(kali@kali)-[~/CIP-B103-Lab3]
$
```

```
(kali@kali)-[~/CIP-B103-Lab3]
$ ls -lh evidence/bounded_syn_activity.pcapng
-rw-r--r-- 1 kali kali 1.4K Jul 26 11:34 evidence/bounded_syn_activity.pcapng

(kali@kali)-[~/CIP-B103-Lab3]
$ cd ~/CIP-B103-Lab3

(kali@kali)-[~/CIP-B103-Lab3]
$ mkdir -p working

(kali@kali)-[~/CIP-B103-Lab3]
$ cp --preserve=timestamps evidence/bounded_syn_activity.pcapng working/bounded_syn_activity_working.pcapng

(kali@kali)-[~/CIP-B103-Lab3]
$ ls -lh working/
total 4.0K
-rw-r--r-- 1 kali kali 1.4K Jul 26 11:34 bounded_syn_activity_working.pcapng

(kali@kali)-[~/CIP-B103-Lab3]
$ sha256sum evidence/bounded_syn_activity.pcapng working/bounded_syn_activity_working.pcapng | tee reports/bounded_capture_hashes.txt
695f2e51f86fc56e493e09caef1badfc189f42a609204fb38dea63bb1f1ae944 evidence/bounded_syn_activity.pcapng
695f2e51f86fc56e493e09caef1badfc189f42a609204fb38dea63bb1f1ae944 working/bounded_syn_activity_working.pcapng
```

```
(kali@kali)-[~/CIP-B103-Lab3]
$ tshark -r working/bounded_syn_activity_working.pcapng \
-Y 'tcp.flags.syn==1 && tcp.flags.ack==0' \
-T fields \
-e frame.number \
-e frame.time_epoch \
-e ip.src \
-e tcp.srcport \
-e ip.dst \
-e tcp.dstport \
-e tcp.seq \
| tee reports/initial_syns.tsv
1      1785080029.631886812      127.0.0.1      51631      127.0.0.1      80      0
4      1785080029.632391919      127.0.0.1      45323      127.0.0.1      80      0
7      1785080029.632815713      127.0.0.1      9145       127.0.0.1      80      0
10     1785080029.633296510      127.0.0.1      33581      127.0.0.1      80      0

(kali@kali)-[~/CIP-B103-Lab3]
$ tshark -r working/bounded_syn_activity_working.pcapng \
-Y 'tcp.flags.syn==1 && tcp.flags.ack==1' \
-T fields \
-e frame.number \
-e frame.time_epoch \
-e ip.src \
-e tcp.srcport \
-e ip.dst \
```

```

-e tcp.dstport \
-e tcp.ack \
| tee reports/syn_ack_responses.tsv
2      1785080029.631907523    127.0.0.1      80      127.0.0.1      51631    1
5      1785080029.632400264    127.0.0.1      80      127.0.0.1      45323    1
8      1785080029.632823527    127.0.0.1      80      127.0.0.1      9145     1
11     1785080029.633305385    127.0.0.1      80      127.0.0.1      33581    1

(kali㉿kali)-[~/CIP-B103-Lab3]
$ tshark -r working/bounded_syn_activity_working.pcapng \
-Y 'tcp.flags.reset=1 || (tcp.flags.ack=1 && tcp.len=0)' \
-T fields \
-e frame.number \
-e frame.time_epoch \
-e ip.src \
-e tcp.srcport \
-e ip.dst \
-e tcp.dstport \
-e tcp.flags \
| tee reports/ack_reset_candidates.tsv
2      1785080029.631907523    127.0.0.1      80      127.0.0.1      51631    0x0012
3      1785080029.631913117    127.0.0.1      51631   127.0.0.1      80       0x0004
5      1785080029.632400264    127.0.0.1      80      127.0.0.1      45323    0x0012
6      1785080029.632403484    127.0.0.1      45323   127.0.0.1      80       0x0004
8      1785080029.632823527    127.0.0.1      80      127.0.0.1      9145     0x0012
9      1785080029.632826416    127.0.0.1      9145    127.0.0.1      80       0x0004
11     1785080029.633305385    127.0.0.1      80      127.0.0.1      33581    0x0012

```

```

(kali㉿kali)-[~/CIP-B103-Lab3]
$ cd ~/CIP-B103-Lab3

PCAP=working/bounded_syn_activity_working.pcapng

(kali㉿kali)-[~/CIP-B103-Lab3]
$ echo $PCAP
working/bounded_syn_activity_working.pcapng

(kali㉿kali)-[~/CIP-B103-Lab3]
$ tshark -r "$PCAP" -Y 'tcp.flags.syn=1 && tcp.flags.ack=0' -T fields \
-e ip.src -e ip.dst -e tcp.dstport | sort | uniq -c | sort -nr \
| tee reports/syn_counts_by_pair.txt
4 127.0.0.1      127.0.0.1      80

(kali㉿kali)-[~/CIP-B103-Lab3]
$ tshark -r "$PCAP" -Y 'tcp.flags.syn=1 && tcp.flags.ack=0' -T fields \
-e tcp.srcport \
| sort -n | uniq | tee reports/unique_syn_source_ports.txt
9145
33581
45323
51631

```

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ tshark -r "$PCAP" -q -z expert | tee reports/expert_info.txt
```

Warns (4)

Frequency	Group	Protocol	Summary
4	Sequence	TCP	Connection reset (RST)

Notes (4)

Frequency	Group	Protocol	Summary
4	Protocol	TCP	The SYN packet does not contain a SACK PERM option

Chats (8)

Frequency	Group	Protocol	Summary
4	Sequence	TCP	Connection establish request (SYN): server port 80
4	Sequence	TCP	Connection establish acknowledge (SYN+ACK): server port 80

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ tshark -r "$PCAP" \
-Y 'tcp.analysis.retransmission || tcp.analysis.lost_segment || tcp.analysis.duplicate_ack' \
-T fields \
-e frame.number \
-e frame.time \
-e _ws.col.Info \
```

```
(kali㉿kali)-[~/CIP-B103-Lab3]
$ tree ~/CIP-B103-Lab3
/home/kali/CIP-B103-Lab3
├── evidence
│   ├── bounded_syn_activity.pcapng
│   ├── mySYNFloodCapture.pcap
│   └── normal_http.pcapng
├── exported
├── reports
│   ├── ack_reset_candidates.tsv
│   ├── bounded_capture_hashes.txt
│   ├── bounded_syn_packets.tsv
│   ├── expert_info.txt
│   ├── initial_syms.tsv
│   ├── normal_handshake_flags.tsv
│   ├── syn_ack_responses.tsv
│   ├── syn_capture_sha256.txt
│   ├── syn_counts_by_pair.txt
│   ├── tcp_analysis_events.tsv
│   └── unique_syn_source_ports.txt
├── screenshots
├── scripts
│   └── syn_probe_lab.py
├── working
│   └── bounded_syn_activity_working.pcapng
```

7 directories, 16 files